Free balancing equations worksheet worksheet 26

Maths | 10th grade | Algebra

1. If 2x - 3 = 4, what is the value of x?

A) 1
B) 3.5
C) 3.5
D) 4

2. Simplify the expression: 3(a + 2) - 2(a - 4)

A) a + 14
B) a + 10
C) 5a + 2
D) a + 8

3. Solve for y: 5(y - 1) = 4y + 3

A) 8
B) 9
C) 5
D) 4

4. The sum of three consecutive even numbers is 54. What is the smallest number?

A) 16
B) 18
C) 20
D) 22

5. What is the solution to the quadratic equation x^2 - 5x + 6 = 0?

A) x = 2 or x = 3
B) x = -2 or x = -3
C) x = 1 or x = 6
D) x = 2 or x = 6

6. Factor the expression completely: x^2 - 9x + 18

A) (x-6)(x-3)
B) (x+6)(x-3)
C) (x-9)(x+2)
D) (x+3)(x-6)

7. The function f(x) = 2x^2 - 8x + 6 opens in which direction?

A) Upward
B) Downward
C) Left
D) Right

8. Solve the inequality: 3x + 2 < 11

A) x < 3
B) x < 4

C) $x > 3$
D) $x > 4$

9. If $f(x) = 4x - 5$, what is $f(3)$?

A) 7
B) 12
C) 13
D) 15

10. The slope of the line passing through points (2, 3) and (6, -1) is:

A) -1
B) -2
C) 0.5
D) 2

Answer Key

1. B
Add 3 to both sides to get 2x = 7, then divide by 2 to find x = 3.5.

2. A
Distribute to get 3a + 6 - 2a + 8, then combine like terms to simplify to a + 14.

3. D
Distribute to get 5y - 5 = 4y + 3, then subtract 4y and add 5 to get y = 8.

4. A
Let the numbers be x, x+2, x+4. The equation is x + x + 2 + x + 4 = 54. Solving gives x = 16.

5. A
Factor the equation to (x-2)(x-3)=0, giving solutions x = 2 and x = 3.

6. A
Look for two numbers that multiply to 18 and add to -9, which are -6 and -3.

7. A
The quadratic function opens upward because the coefficient of x^2 is positive.

8. A
Subtract 2 to get 3x < 9, then divide by 3 to find x < 3.

9. C
Substitute 3 into the function: f(3) = 4(3) - 5 = 12 - 5 = 7.

10. B
Use the slope formula: m = (-1 - 3) / (6 - 2) = -4 / 4 = -1.